

VISHAL SHISHODIA

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CAREER OBJECTIVE

B.Tech CSE (AI & ML) undergraduate at IMSEC-AKTU, proficient in Python, PyTorch, Scikit-learn, and LLM integration, with hands-on project experience in end-to-end ML systems — seeking an AI/ML internship to apply skills in model development, data pipelines, and intelligent application design.

TECHNICAL SKILLS

Programming Languages: Python, Java, SQL

Machine Learning: Supervised & Unsupervised Learning, Regression, Classification, Clustering, Feature Engineering

Deep Learning: Neural Networks, CNNs, Backpropagation, Model Training & Evaluation

ML Frameworks & Libraries: PyTorch, Scikit-learn, NumPy, Pandas, Matplotlib, Seaborn

LLM & Generative AI: LLM Integration, Prompt Engineering, Ollama (LLaMA 3), RAG concepts

Backend & DevOps: FastAPI, REST API Design, Docker, Docker Compose

Data Skills: EDA, Data Cleaning, Preprocessing, Cross-Validation, Precision/Recall/F1

Core CS Concepts: Data Structures & Algorithms, Object-Oriented Programming, DBMS

Databases: MySQL, SQL

PROJECTS

AI Code Review & Optimization Engine | <https://github.com/vishishodia/code-review-ai>

Technologies: Python, FastAPI, LLaMA 3 (Ollama), Docker, Docker Compose, REST API

- Built a static analysis engine to detect inefficient code patterns (nested loops, unbounded recursion) and estimate time complexity from $O(1)$ to $O(n^3)$ using rule-based analysis across modular analyzer components.
- Integrated LLaMA 3 via Ollama locally for context-aware code optimization suggestions; architected a modular FastAPI backend separating analysis, complexity estimation, and AI suggestion layers.

Smart India Harvest – Crop Yield Prediction System

Technologies: Python, Scikit-learn, Pandas, NumPy, Regression Modeling

- Developed an end-to-end ML pipeline — data ingestion, preprocessing, feature engineering, model training, and evaluation — to predict crop yield from 1,000+ agricultural records using regression models.
- Reduced manual data processing effort by approximately 40% through automated preprocessing and feature selection; achieved strong cross-validation performance on held-out test splits.

Urban Air Quality Analysis & Prediction System | <https://github.com/vishishodia/AirQualityProject>

Technologies: Python, Scikit-learn, Matplotlib, Seaborn, Supervised ML, EDA

- Performed comprehensive EDA on PM2.5 and NO₂ datasets using Matplotlib and Seaborn to uncover pollution trends, seasonal patterns, and feature correlations.
- Trained supervised ML classification and regression models to forecast Air Quality Index (AQI) levels, enabling data-driven environmental insights and early-warning predictions.

EDUCATION

Bachelor of Technology (B.Tech) – Computer Science & Engineering (AI & ML) 2023 – 2027

IMS Engineering College (IMSEC), Affiliated to AKTU | CGPA: 7.0

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, DBMS, Machine Learning, Probability & Statistics, Software Engineering

CERTIFICATIONS & ACHIEVEMENTS

- **Google for Developers & EduSkills (AICTE Initiative)** — AI-ML Virtual Internship
- **Anthropic** — Claude-101 Certification
- **Core AI/ML concepts and applications** — AI Fluency: Framework and Foundations
- **Consistent daily problem-solving streak** — GeeksforGeeks 160-Day Coding Challenge